

Your Solar Assessment Report

A personalised rooftop solar analysis for your address

Location-specific solar, cost, subsidy, and savings estimates



YOUR SOLAR SCORE

Excellent

Composite · Uttar Pradesh

Your rooftop has excellent solar potential. Highly suitable for rooftop solar installation.

PREPARED FOR

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LOCATION COORDINATES

28.3111629°N, 78.8693491°E

REPORT ID

SS-20260825-4C86B16D4C

LOCATION

Bisauli, Uttar Pradesh

GENERATED ON

25 August 2026, 02:15 PM IST

REPORT ID

SS-20260825-4C86B16D4C

DATA SOURCE

Climate zones · ALMM

AT A GLANCE

- Recommended system: **3.96 kWp** (9 panels)
- Estimated annual generation: **4,886 kWh**
- Net cost after subsidy: **₹1,20,000**
- Estimated payback: **4.6 years**

About This Report

This report estimates what a rooftop solar system on this specific building would produce, what it would cost after subsidy, and what it would save. It covers all 28 states and 8 union territories, using each state's published residential tariff and the PM Surya Ghar subsidy slabs in force on the date shown.

Methodology: Method: long-term satellite solar irradiance for these coordinates, the state's published residential tariff and net-metering rate, the MNRE ALMM approved module list, and the PM Surya Ghar subsidy slabs. Roof area and shading come from the answers given during the assessment.

Disclaimer: An indicative assessment based on satellite data and standard engineering assumptions. Actual performance varies with installation quality, equipment and usage. Confirm subsidy eligibility and tariff rates with your DISCOM before committing.

SECTION 2

Solar Score & Rooftop Analysis

SOLAR SCORE



Excellent

Composite

Your rooftop has excellent solar potential. Highly suitable for rooftop solar installation.

CLIMATE ZONE

Composite

Mixed climate, moderate radiation

Annual Solar Radiation **1825 kWh/m²**

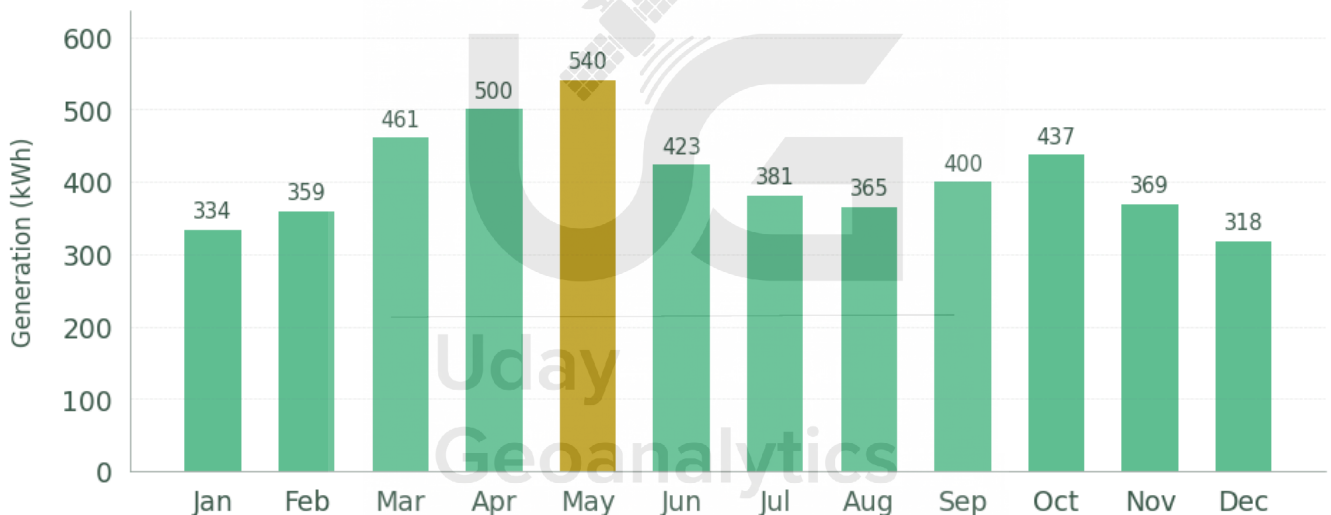
Peak Sun Hours **5.22 hrs/day**

Best Solar Months **May, Apr, Mar**

Least Generation **Dec, Jan**

Estimated Monthly Solar Generation

Based on Composite solar irradiance for Bisauli



25°C

AVG TEMPERATURE

230-260

CLEAR SKY DAYS/YEAR

May, Apr, Mar

BEST MONTHS

ROOFTOP ANALYSIS

130

sq.m

TOTAL AREA

73.9

sq.m

USABLE AREA

0.821

SHADOW FACTOR

flat_rcc

ROOF TYPE

Assessment: This rooftop is well suited to solar. Irradiance, usable area and your state's tariff all work in your favour. Clean the panels every two to three months, and after any dust storm. Soiling is the largest avoidable loss in Indian conditions.

SECTION 3

Recommended Solar System

RECOMMENDED SYSTEM SIZE

3.96
kWp System

9 panels × 440W · South-facing at 28° tilt

CLIMATE SUITABILITY

Composite

Mixed climate, moderate radiation Mono PERC panels perform optimally in this climate.

RECOMMENDED PANEL

ALMM Certified

Adani Solar

ASM-440M-GG5

Panel Type	Mono PERC
Wattage	440 Wp
Efficiency	64.7%
Warranty	25 Years Linear

Why this panel: Chosen for its efficiency in this climate zone, its ALMM listing, and the best output achievable within the usable area available.

SYSTEM DESIGN

System Capacity	3.96 kWp
Number of Panels	9
Optimal Tilt	28°
Azimuth	180° (South)
Total Panel Area	17.3 sq.m
Performance Ratio	0.6741
System Efficiency	64.7%

ESTIMATED ENERGY GENERATION

13.39

kWh

DAILY AVERAGE

407

kWh

MONTHLY AVERAGE

4,886

kWh

ANNUAL TOTAL

Estimates are based on Climate-zone averages (NASA POWER unavailable), 5 kWh/m²/day irradiance, and 0.6741 performance ratio. Actual output may vary by about 10 percent either way depending on installation quality and weather in a given year.

SYSTEM COMPONENTS OVERVIEW

COMPONENT	SPECIFICATION
Solar Panels	9 × 440W Mono PERC
Total Capacity	3.96 kWp
Inverter Size	3.96 kW (String Inverter)
Panel Area Required	17.3 sq.m
Mounting Type	Rooftop Fixed Tilt (28° South)
Performance Ratio	0.6741
Expected Lifespan	25 Years (25 Years Linear panel warranty)

SECTION 4

Investment & Subsidy Breakdown

COST BREAKDOWN

Gross System Cost	₹1,98,000
PM Surya Ghar Subsidy	- ₹78,000
State Subsidy	- ₹0
Total Subsidy	₹78,000

YOUR NET COST AFTER SUBSIDY

₹1,20,000

Subsidy covers 39.4% of system cost

YOUR DISCOM & BILLING

DISCOM	UPPCL / DVVNL
Tariff	₹6/kWh
Net Metering	Available

SOLAR LOAN OPTION

Interest Rate	6.75% p.a.
Loan Tenure	7 Years
Collateral	None required

MONTHLY EMI

₹1,796

Often lower than current electricity bill

PM SURYA GHAR SUBSIDY SLABS

Up to 2 kW	₹60,000
2 kW – 3 kW	₹78,000
Above 3 kW	₹78,000 (capped)

WHAT THIS MEANS FOR YOU

- Upfront cost reduced by **39.4%** through PM Surya Ghar government subsidy — no complex paperwork required
- Monthly EMI of **₹1,796** is often lower than your current electricity bill — you start saving from Day 1
- Net metering available via UPPCL / DVVNL — export surplus power to the grid and earn credits on your bill
- Based on your consumption, this system covers about **76%** of the electricity you currently buy — the rest continues to come from the grid
- System payback in just **4.6 years** — then 20+ years of near-free electricity generation
- Protection from electricity tariff hikes — with solar, your effective cost per unit stays fixed for 25 years
- 25-year total savings of **₹13,02,111** — a long-term wealth building asset on your rooftop
- ALMM-certified panels carry **25 Years Linear** performance guarantee — fully backed by MNRE quality standards
- Your investment contributes to India's 500 GW renewable target and directly reduces national grid carbon emissions

SECTION 5

Financial Returns & Savings

₹2,443

MONTHLY SAVINGS

4.6

Years
PAYBACK PERIOD

₹13,02,111

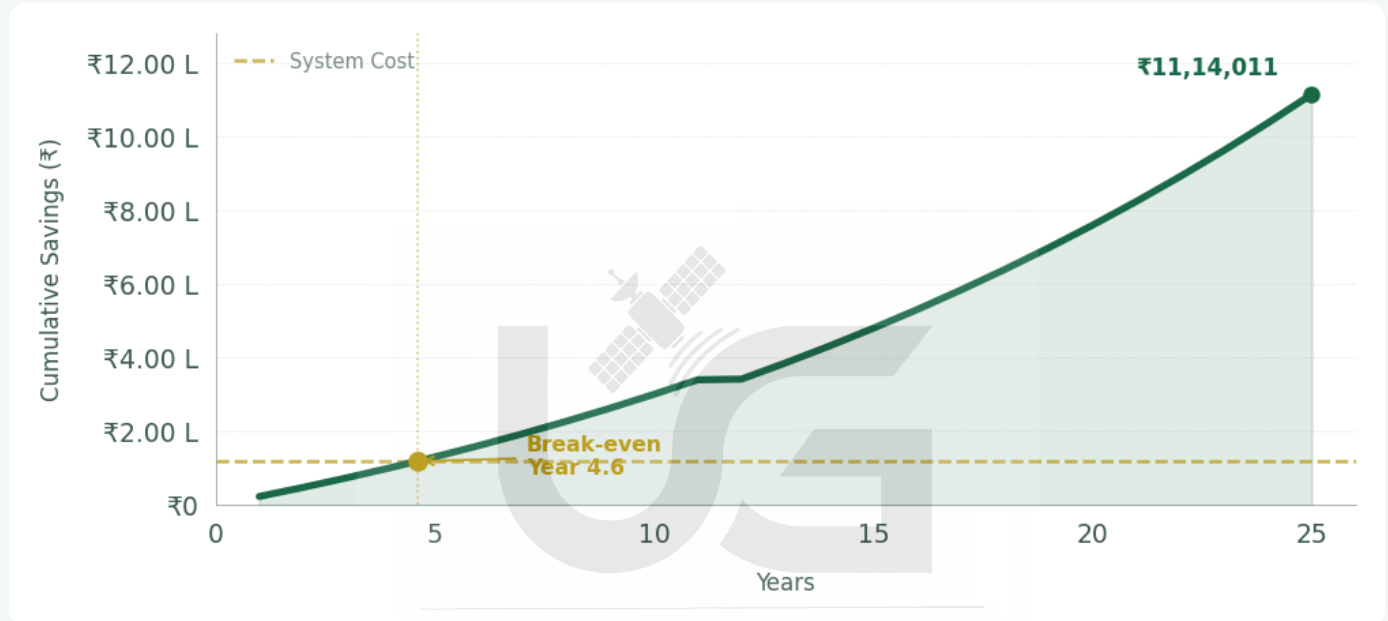
25-YEAR SAVINGS

828.3%

ROI

Your Solar Savings Growth Over 25 Years

Cumulative savings vs system cost · Break-even at Year 4.6



Your system pays for itself in 4.6 years and generates total savings beyond ₹13,02,111 over 25 years.

YEAR-WISE SAVINGS BREAKDOWN

Year	Annual Savings	Cumulative
Year 1	₹29,318	₹23,378
Year 3	₹32,000	₹74,128
Year 5	₹34,929	₹1,30,609
Year 10	₹43,475	₹3,00,445
Year 15	₹54,113	₹4,79,505
Year 20	₹67,354	₹7,58,937
Year 25	₹83,836	₹11,14,011

Savings assume 5% annual tariff escalation, 0.6741 performance ratio, 0.5% annual degradation.

SECTION 6

Environmental Impact

Your contribution to a cleaner, greener future

YEARLY ENVIRONMENTAL IMPACT

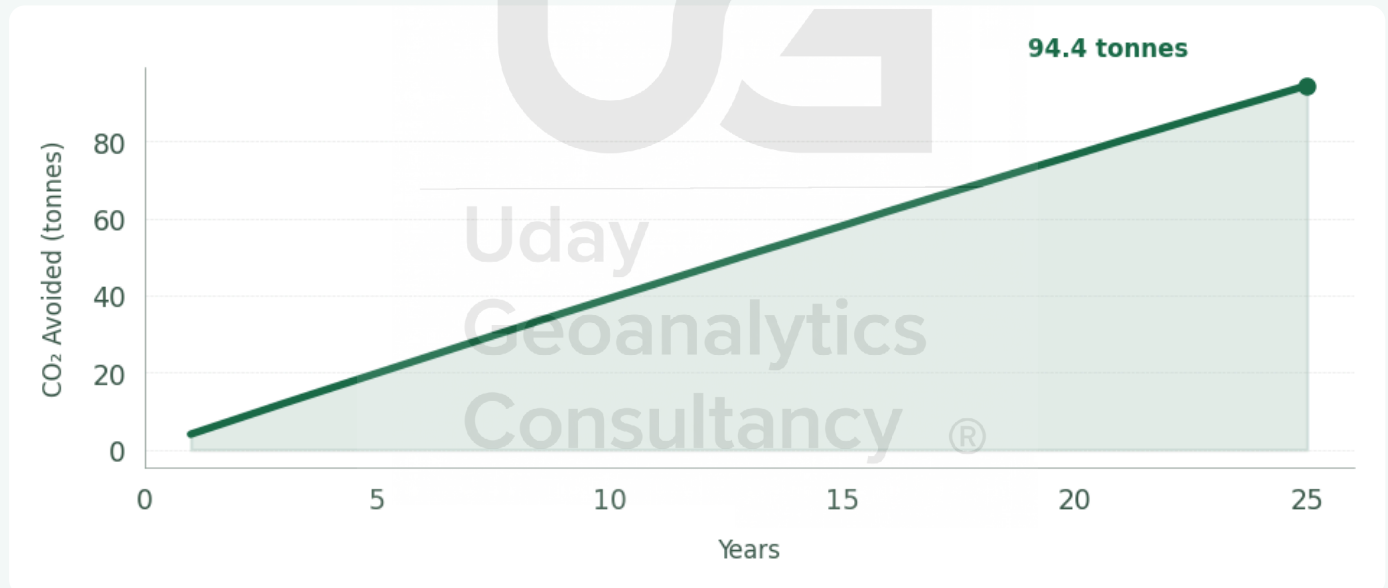


YOUR LIFETIME ENVIRONMENTAL CONTRIBUTION



CO₂ Savings Growth Over 25 Years

Cumulative carbon offset from your solar system



EMISSION FACTORS USED

Electricity Emission Factor	0.82 kg CO ₂ /kWh
Source	CEA India Grid 2024
Coal Equivalence	0.4 kg/kWh
Tree Absorption Rate	20 kg CO ₂ /tree/year

Used to estimate carbon savings based on national grid emission data.

INDIA'S CLIMATE GOALS

India targets 500 GW renewable energy capacity by 2030 and net-zero emissions by 2070.

- 50% electricity from renewables by 2030
- 1 billion tonnes CO₂ reduction by 2030
- Net zero carbon emissions by 2070

Your system contributes directly to these national targets.

"By installing this solar system, you will offset 100.2 tonnes of CO₂ over 25 years — equivalent to planting 5000 trees. You're not just saving money, you're building a cleaner future."

SECTION 7

Your Next Steps

Follow these steps to install your rooftop solar system smoothly.

- 1 **Register on PM Surya Ghar Portal**
Visit pmsuryaghar.gov.in and register with your electricity consumer number and Aadhaar.
10-15 minutes
- 2 **Apply to Your DISCOM**
Submit net metering application to UPPCL / DVVNL. They will verify your connection and approve feasibility.
3-7 working days
- 3 **Choose an Empanelled Installer**
Select an MNRE-empanelled installer from the portal. Compare quotes for your 3.96 kWp system.
1-3 days
- 4 **Arrange Financing If Needed**
Apply for solar loan at 6.75% p.a. EMI: ₹1,796/month. Many banks offer pre-approved solar loans.
2-5 working days
- 5 **Installation and Commissioning**
Installer sets up 9 panels (Adani Solar ASM-440M-GG5) on your rooftop with inverter and wiring.
2-3 days
- 6 **Net Meter Installation**
DISCOM installs bi-directional net meter. After inspection, your system goes live on the grid.
3-7 working days
- 7 **Start Saving**
Your system generates **13.39 kWh/day**. Monthly savings of **₹2,443** start from Day 1. Subsidy is credited directly to your bank account.

YOUR SOLAR SNAPSHOT

System Size	3.96 kWp
Net Cost	₹1,20,000
Monthly Savings	₹2,443
Payback Period	4.6 Years
25-Year Savings	₹13,02,111
CO ₂ Saved/Year	4,007 kg

NEED HELP? OFFICIAL RESOURCES

PM Surya Ghar Portal

pmsuryaghar.gov.in

MNRE Helpline

1800-180-3333 (Toll Free)

Your DISCOM

UPPCL / DVVNL

SolarSight Support

solarsight.in · ugaconsult.com

Contact@ugaconsult.com · +91 75794 95251

Ward No 5, Andhratharhi, Madhubani, Bihar 847401

CIN U63111BR2026PTC086044

Report Validity: This report is valid for 90 days.
Tariffs and subsidy slabs change; run a fresh assessment after that.



Scan for more details at
solarsight.in

Start your solar journey

Apply on the national portal and begin saving from the first month of generation.

Visit pmsuryaghar.gov.in